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01 Introduction HKA: Integration of Digital Twins (AAS) and CAD (Siemens NX)

Within the scope of this development project, the informational and conceptual bridge between the Asset Administration Shell (AAS) and the CAD system Siemens NX is built.

> "The Asset Administration Shell is the standardized digital representation of an asset. It structures all relevant data and properties of a physical or logical object throughout its entire lifecycle, thus forming the technical foundation for the Digital Twin in Industry 4.0." [1]

The goal is to enable working with Digital Twins across system boundaries and to ensure continuous data utilization throughout the entire product lifecycle.

Methodologically, the project is divided into two consecutive phases:

Phase 1: Fundamentals and Application Scenarios in the Product Lifecycle

In the first phase of the project, the theoretical fundamentals of the AAS were developed, and the specific added value and application scenarios offered by connecting a CAD system to the administration shell were evaluated.

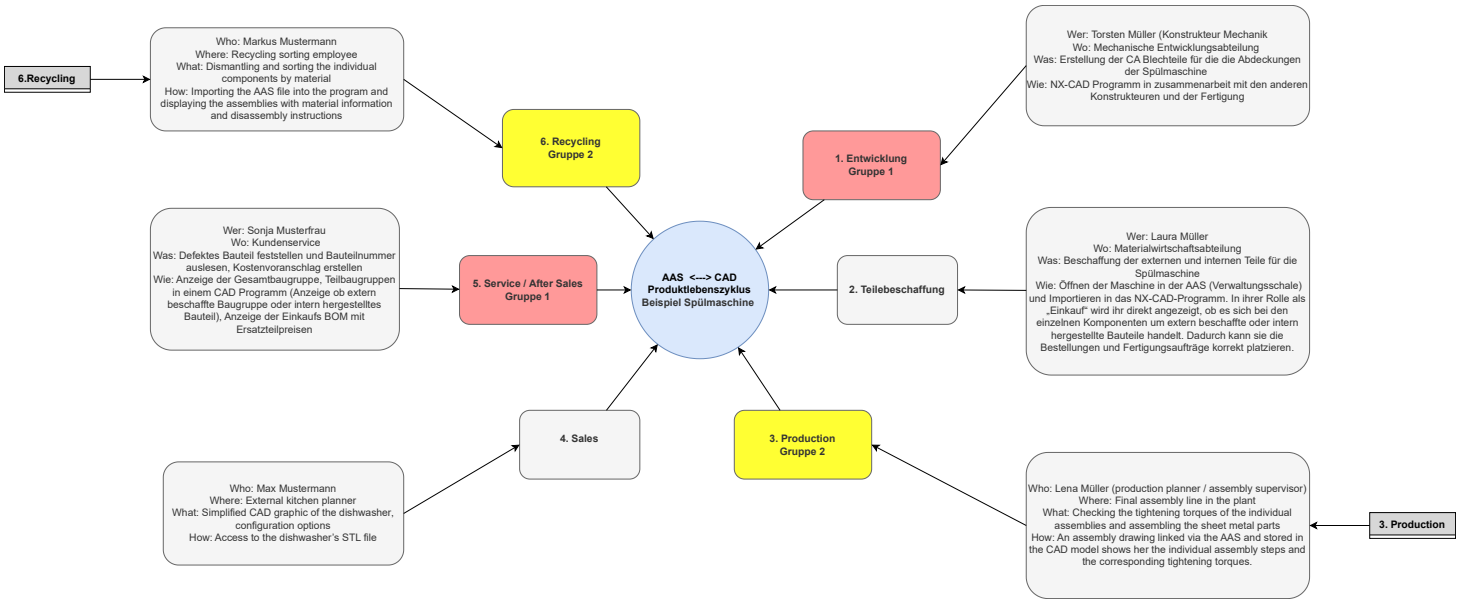
A practical example of a **dishwasher** serves to illustrate the concepts. Along its product lifecycle, highly varying requirements arise in each phase regarding the provision, updating, and structuring of data from the Digital Twin.

The considered lifecycle encompasses the following six phases:

1. Development (CAD & Engineering)
2. Parts Procurement
3. **Production (Manufacturing)**
4. Sales
5. Service / After-Sales

6. **Recycling (End-of-Life)**

*As shown in the following diagram, these phases form a continuous cycle. Within the scope of this project, the **Production** and **Recycling** phases, in particular, were elaborated in depth. The detailed elaboration of these use cases is the subject of the following chapters.*



Phase 2: Technical Implementation and Interface Functions

Based on the conceptual preliminary work of the first phase, the focus of the second project phase was on the technical implementation. Here, the central core functions required for practical work with AASX data packages and the CAD program Siemens NX were identified and developed.

> **Documentation Note ("Living Document"):**
> Since research and work on this repository will continue in the future, this documentation is designed directly as the structure for the GitHub repository (`README.md` and linked Markdown files). Redundant, separately maintained documentation was deliberately avoided. This ensures that code changes and documentation always remain synchronous, transparently versioned, and accessible in a central location for future developers.

3. Scenario: Production (Manufacturing)

3.1 Detailed description of the scenario

Lena Müller works as a production planner and assembly manager in final assembly. She is responsible for the correct execution of the assembly processes as well as for the quality assurance of the assembled modules. Lena checks the tightening torques of the individual screw connections and assembles the corresponding sheet metal components. In doing so, she ensures that all assembly specifications are strictly adhered to in order to guarantee the functionality, safety, and quality of the product.

The relevant assembly information is provided to Lena digitally. An assembly drawing, linked to the respective physical asset via the Asset Administration Shell (AAS), is stored in the CAD model. Using a digital device (e.g., an industrial tablet or HMI), Lena can access this information in context.

3.2 Process flow description

- **Product / Asset Identification:** Lena identifies the product to be assembled using a serial number or a QR code.
- **Calling up the AAS:** The corresponding Asset Administration Shell is loaded via the manufacturing system.
- **Accessing assembly information:** The AAS points to the linked CAD model and the corresponding assembly drawing.
- **Display of assembly steps:** The individual assembly steps are presented sequentially, including the required tools.
- **Checking tightening torques:** The target torques for each screw connection are displayed and verified with the torque tool in use.
- **Assembly of sheet metal components:** Lena assembles the sheet metal components according to the drawing and the defined process steps.
- **Documentation and feedback:** The completed assembly and the verified torques are documented in the system and, if necessary, reported back to higher-level systems.

3.3 Required files and resources

- **CAD models** (e.g., STEP, JT, native CAD formats)
- **Assembly drawings** (2D/3D, e.g., PDF or directly in CAD)
- **Torque specifications** (e.g., tables or metadata)
- **Asset Administration Shell** (AAS model in JSON or XML format)
- **Digital devices** (tablet, industrial PC, HMI)

- **Torque tools** (ideally with a digital interface)
- **MES/PLM systems** for data management

3.4 Technical implementation

3.4.1 Implementation in the Asset Administration Shell (AAS)

The Asset Administration Shell serves as a central data hub for assembly- and quality-relevant information. Technically, the implementation is carried out via clearly defined `SubmodelElements` that standardize assembly knowledge, CAD references, and torque specifications.

Assembly and torque data

- **Submodel "AssemblyInformation":** Contains the complete description of the assembly steps as structured `SubmodelElements` (e.g., `Operation`, `Sequence`, `StepNumber`).
- **Tightening torques:** Each screw connection is described by a `Property` element:
  - `FastenerID`
  - `TorqueNominal` (e.g., 12 Nm)
  - `TorqueTolerance` (e.g., ±10%)
  - `ToolType` (e.g., electronic torque wrench)

> **Example:** The screw connection of the side panel has the attribute `TorqueNominal = 12 Nm`, `TorqueTolerance = ±1.2 Nm`.

3.4.2 Implementation in the CAD environment

The CAD environment (viewer or assembly frontend at Lena's workstation) acts as a visual and interactive interface between the worker and the digital product twin.

- Each component and screw connection in the CAD model is linked to the `InstanceId` of the corresponding AAS component.
- If Lena clicks on a screw in the 3D model, the **target torque**, **tolerance**, and **tool type** are loaded directly from the AAS and displayed.

Feedback and reporting channel (As-Planned vs. As-Built)

Lena can report deviations directly in the CAD interface, e.g.:

- Screw is inaccessible
- Component does not fit stress-free
- Torque cannot be achieved structurally

These reports are stored as an `Event` or `Annotation` directly in the AAS and are made available to engineering and production planning.

6. Scenario: Recycling (End-of-Life)

6.1 Detailed description of the scenario

In 2026, the linear economy ("take, make, dispose") was legally ended by the *Ecodesign for Sustainable Products Regulation* (ESPR). Every dishwasher sold in the EU must now have a Digital Product Passport (DPP), which is technically implemented as an Asset Administration Shell (AAS).

Markus Mustermann, a specialized employee in a certified dismantling center, faces a challenge: A modern dishwasher consists of over 150 different materials, including valuable stainless steel, copper-bearing motors, electronics with rare earths, and problematic bitumen insulation mats.

Without the AAS, the dishwasher would be a "black box" for Markus. He would have to shred it, which severely degrades material purity. Thanks to the AAS, Markus becomes an Urban Miner: He accesses the digital twin, which tells him not only what is installed but also how to separate it non-destructively to maximize recycling yields.

6.2 Process flow description

The workflow is divided into five phases:

1. **Capture & Initialization:** Markus scans the QR code lasered onto the housing frame. The AAS opens the corresponding model for this.
2. **Status & Safety Analysis:** Before Markus uses any tools, the system checks the "Safety" submodel. It displays whether capacitors need to be discharged or if chemical residues (e.g., rinse aid remnants) are present in the hoses.
3. **Virtual Exploration:** Markus receives the dismantling drawings in the AAS and disassembles the machine based on them. He is shown which part is made of which material. At the end, he enters how much mass of each material he has separated by type.
4. **Guided Dismantling:** Markus follows the steps stored in the AAS. The system dictates:
    - *Step 1:* Loosen the Torx screws (T15) on the back panel.
    - *Step 2:* Disconnect the wiring harness.
    - *Step 3:* Remove the insulation.

6.3 Required files and resources

The following items are required for the implementation:

- **The AASX container file:** Contains the logical structure and all references.
- **Identification:** Unique ID.
- **Digital Product Passport (DPP):** Contains the legally required ESPR data (carbon footprint, recyclate content).
- **Bill of Materials (R-BOM):** The hierarchical list as well as properties for recycling all components.
- **TechnicalProperties:** Material data sheets per component, provided by the respective manufacturers and merged into the dishwasher's AAS.
- **CAD resources:** Dismantling drawings and 3D model for disassembly.

6.4 Technical implementation

6.4.1 Implementation in the Asset Administration Shell (AAS)

The AAS serves as a data hub. All installed components have their properties stored by the respective manufacturers, and these are assembled into a dishwasher.

- **Material Identification:** Every component in the dishwasher receives a `Property` element with the material ID.
  - *Example:* The cladding elements receive the attribute `Material: V2A (Stainless Steel)`.
- **API Interface:** The AAS provides a REST API through which the recycling software makes queries (e.g., `GET /submodels/MaterialComposition/submodel-elements/CopperWeight`).

6.4.2 Implementation in the CAD environment

The CAD environment serves as the central data source for creating the Asset Administration Shell (AAS). The process is structured as follows:

- **Provision of documents:** Dismantling drawings in PDF format are generated directly from the CAD system and made available to the AAS.
- **Material flow analysis:** An automated list is generated that records all components, including their volumes and materials. From this, the exact masses of the different raw materials can be derived.
- **Target-actual comparison in the recycling process:** As part of the recycling process, this data is calculated for each dishwasher variant and stored in the AAS as target values.
- **Proof of obligation:** After dismantling, the recycler is obliged to enter the actually recovered masses of the different materials as actual values into the AAS.

- **Feedback loop to the manufacturer:** This data is fed back to the manufacturer. This gives the manufacturer seamless proof of the percentage to which the machine was sorted and dismantled.
- **Visualization:** Given high system complexity, an additional 3D model can be provided to the recycler.
- **Additional visualization:** The system calculates the recycling priority based on the stored material properties. Components containing pollutants are displayed flashing red.

6.5 Regulations and Legal Guidelines (Additional Requirements)

1. **ESPR (Ecodesign for Sustainable Products Regulation)**
    - **Standard tools & dismantlability:** The existing Regulation (EU) 2019/2022 for dishwashers already stipulates that certain parts (such as pumps or motors) must be replaceable with "commonly available tools" and without causing permanent damage to the appliance.
    - **Dismantling time:** The new ESPR (Regulation (EU) 2024/1781) allows the EU to set explicit performance requirements in product-specific "delegated acts". For the revision of the dishwasher rules (planned from 2026), discussions are underway to include dismantling time as a metric for recyclability. [2], [3]
2. **EU Battery Regulation (2023/1542)**
    - **Removability:** From February 18, 2027, portable batteries must be designed so that they can be easily removed and replaced by the end user or qualified personnel.
    - **Battery passport:** While the full "battery passport" primarily applies to industrial (>2 kWh) and traction batteries, information on chemical composition and safe removal for small batteries must be stored in the Digital Product Passport (DPP) of the main device (here, the dishwasher's AAS). [4]
3. **WEEE Directive (Waste Electrical and Electronic Equipment)**
    - WEEE regulates the handling of electronic waste after use.
    - **Collection and recovery targets:** For large appliances (Category 1), which include dishwashers, the directive prescribes a recovery target of 85% and a target for preparation for reuse and recycling of 80%.
    - **Role of the AAS:** The AAS is the tool to guarantee "material hygiene". These high targets can only be achieved economically through single-origin separation (e.g., grade 1.4301 stainless steel instead of mixed scrap). [5]
4. **Substances of Concern (REACH & SCIP)**
    - **Information obligation:** Since January 2021, manufacturers must submit information on Substances of Very High Concern (SVHC) to the ECHA SCIP database if the concentration exceeds 0.1%.

- **AAS Integration:** The AAS serves as an interface to make this complex chemical data directly visualizable for Markus Mustermann at his workstation (e.g., warning label for bitumen mats or flame retardants in circuit boards), without him having to search manually in databases. [6], [7]

02 Introduction GitHub

AAS-CAD Integration: Connection between Asset Administration Shell and CAD

Introductory document for our AAS-CAD Integration project.

In this project, we are working on enabling the Asset Administration Shell (AAS) – the standardized digital twin – to communicate directly with our CAD systems. The goal is to enable bidirectional data exchange and to provide concrete code examples and workflows for this purpose.

What exactly is this about?

The current state of the art is: 3D models reside in CAD, and the digital twin lives completely isolated from it in its own software environment. However, if we want to implement use cases like intelligent recycling or seamless material tracking, these two worlds must talk to each other. This repository collects scripts and plug-ins to build exactly this bridge.

Core Features Overview

We focus on four fundamental functions to establish the interaction between AAS and CAD. In the table, you will find direct links to the respective explanations and implementations for Siemens NX and

PTC Creo, as well as the current development status for Autodesk Fusion 360.

#	Feature	Implementation in Siemens NX	Implementation in PTC Creo	Implementation in Autodesk Fusion 360
1	**Import of 3D data from the AAS** *(Loading the model from the digital twin into CAD)*	NX Part Loader	Creo Doc	🚧 Work in Progress
2	**Assignment of an Asset ID to a part** *(The fundamental link between the 3D model and the AAS instance)*	NX Asset-ID assign	Creo Doc	🚧 Work in Progress
3	**Synchronize properties** *(Exchanging material data, metadata, etc.)*	NX Sync Properties with AAS	Creo Doc	🚧 Work in Progress
4	**Data export from CAD** *(currently as CSV)*	NX to AAS Export	Creo Doc	🚧 Work in Progress

Detailed Description of the Features

How exactly these four functions work in detail and how they are solved on the system side can be found directly in the linked documents from the table above. There you will find detailed explanations and code examples for Siemens NX and PTC Creo.

Contributing

We always welcome support. If you want to contribute your own scripts, have a solution for another CAD system like SolidWorks, or find bugs, feel free to create a Pull Request.

License

The core project "Connecting CAD to Digital Twins" is licensed under the **Apache License 2.0**. Detailed information can be found in the LICENSE.md file.

> **Important note regarding Siemens NX (UI-Styler):** > This project contains UI code generated with the **Siemens NX 2021 UI-Styler** and manually adapted. To compile or run this code, proprietary NX Open libraries (DLLs) are required, which are **not** part of this repository. The rights to these libraries and to the basic framework of the generated code remain with Siemens AG. Please observe the corresponding license terms (EULA) of your Siemens NX installation.

Further legal notices are listed in the NOTICE.md file.

NX AASX to STEP Import Tool

This project contains a Python script that is executed via the **NX Open Python API**.
The tool is used to select an **AASX file (Asset Administration Shell)**, automatically extract a contained **STEP model** from it, and then import it into **Siemens NX**.

The application thus connects an **Asset Administration Shell (AAS)** with a CAD system by directly opening a STEP model referenced within the AAS in NX.

Overview

The **NX Part Loader** provides a simplified workflow for quickly locating and opening parts.

Features:

- Automatically checks for an existing work part

- Creates a new part if none exists
- Dialog-based file name input
- Recursive search for `.prt` files
- Automatically opens the first matching file found
- Displays errors via the NX MessageBox

Basic functionality (importing a STEP model from an AASX file)

To integrate digital twins or Industry 4.0 models into CAD systems, CAD data must be extracted from the Asset Administration Shell and loaded into the CAD environment.

This script automates this workflow:

1. **Start of the NX script**

The user starts the script directly in Siemens NX.
The script acts as a launcher and starts an external Python script.

2. **Selection of an AASX file**

The external script opens a file selection dialog.
The user selects an .aasx file.

3. **Extraction of the STEP model**

The external script searches the AAS for referenced models and extracts a contained STEP file.

4. **Temporary storage**

The STEP file is stored temporarily.

5. **Import into Siemens NX**

The NX script detects the generated STEP file and automatically imports it into NX.

Code structure and individual functions

The code consists of two scripts:

- **NX launcher script** (performs the import in NX)
- **External AAS script** (extracts the STEP model from the AASX)

NX launcher script

This script is executed directly in Siemens NX and controls the entire workflow.

`main()`

The entry point of the NX script.

Functions:

- Opens the NX Listing Window to display status messages
- Starts the external Python script via subprocess
- Checks whether the STEP file was created
- Ensures that an active NX part exists
- Imports the STEP file into NX

`ensure_work_part()`

This function ensures that an active NX part exists.

If no work part is available:

- a new part is created
- any existing old part is deleted
- the new part is set as the work part

This prevents NXOpen functions from being executed without a valid target part.

`import_step_into_nx()`

This function handles the import of the STEP file into NX.

For this purpose, the NX Step242Importer is used.

Importer configuration:

- Geometry optimization is enabled
- Curves, surfaces, and solids are imported
- PMI data is also included
- Import is performed directly from the file system

The import is then executed automatically.

External script (AAS_TO_NX.py)

This script extracts a STEP model from an AASX file.

`main()`

The entry point of the external script.

Process:

1. Opens a file selection dialog
2. The user selects an .aasx file
3. The file is analyzed
4. A STEP model is extracted
5. The STEP file is saved locally

`select_aasx_file()`

This function opens a file dialog using Tkinter.

The user can select an .aasx file.

If no file is selected, the process is terminated.

`extract_step_from_aasx()`

This function searches the AASX file for contained CAD models.

Process:

1. Import of the AASX file
2. Search for Asset Administration Shells
3. Iteration through the models
4. Checking file formats
5. Detection of a STEP file
6. Extraction of the STEP file from the file store
7. Writing the file to the temporary directory

Installation & execution in NX

1. Ensure that Python is installed.
2. Ensure that the NX Python API is available.
3. Place both scripts in your project directory.

Required scripts:

- NX launcher script:
  AAS_TO_NX.py

4. Open Siemens NX.
5. Start the NX script

The script will then automatically start the complete workflow.

Configuration

Currently, several paths are hardcoded in the code.

Examples:

script_path = C:\Users\chris\AAS-Creo-Bridge\src\AAS_TO_NX.py

step_path = C:\Users\chris\AAS-Creo-Bridge\temp_model.step

Troubleshooting (Error handling)

STEP file is not created

Possible causes:

- The AASX file does not contain a STEP model
- The model format is not STEP
- Access to the file store has failed

External script does not start

Possible causes:

- Incorrect Python path
- Wrong script path
- Missing Python dependencies

Import into NX fails

Possible causes:

- STEP file is corrupted
- NX Step Importer not initialized correctly

- No work part available

Possible improvements

Currently, the script searches for all STP files within the AASX file. If multiple STP files are present, this can lead to issues. The script should be extended to select the correct STP file from the AASX based on a specific identifier or naming convention.

NX PartID to AAS

This project includes a Siemens NX Open application (C#) created using the NX Block UI Styler. For detailed instructions and technical background on working with this tool, please refer to the official Siemens manual **"NX Open Programmer's Guide"**, specifically the chapters on the **"Block UI Styler Introduction"** (for dialog creation and callbacks).

The tool enables designers to assign a unique ID (Part ID) to an active NX part via a simple graphical user interface. This ID is stored as a part attribute (`PART_ID`) and is used to link the 3D model with the **Asset Administration Shell (AAS)** for digital twins.

Basic functionality (Adding the Part ID for the AAS)

To make a 3D model usable in Industry 4.0 scenarios, it must be uniquely identifiable. This script addresses this requirement by providing a defined workflow in NX:

1. **User input:** The user starts the script in NX. A dialog box (defined by the `.dlx` file) opens, in which the desired ID for the Asset Administration Shell is entered.
2. **Validation:** The tool verifies whether an ID has been entered and whether a part is opened and active (Work Part) in NX.
3. **Attribute assignment:** The entered string is written directly into the part metadata via the NX Open API. The attribute **`PART_ID`** is created or overwritten.

Once the part is saved and, for example, exported as a STEP file (including metadata), the AAS can read this `PART_ID` and associate the part with a digital twin.

Code structure and the individual functions

The code is based on the standard template of the NX Block UI Styler. Below is a detailed description of the key functions in `add_PartID.cs` :

`Main()`

The entry point of the application. A new instance of the `add_PartID` class is created, and the dialog is displayed on the user's screen via the `Show()` method.

`initialize_cb()`

(Initialize Callback)
This method is executed immediately when the dialog is loaded. It establishes the connection between the C# logic and the graphical user interface (the `.dlx` file). Here, the input field (block type: String) is located in the UI and assigned to the variable `string0` , so it can be accessed later in the process.

`apply_cb()`

(Apply Callback)
This function is triggered when the user clicks **"Apply"**:

- **Read values:** Retrieves the text from the input field (`string0.Value`).
- **Validation checks (If-Statements):**
  - Checks whether the field is empty. If so, an informational message is displayed and the process is stopped.
  - Checks whether an active work part (`workPart`) exists. Without a part, no attribute can be assigned.
- **Write operation:** The command `workPart.SetAttribute("PART_ID", eingegebeneID);` writes the actual attribute to the NX part.
- **Session Undo:** Creates a visible undo step ("Part ID set"), to keep the NX session clean.

`ok_cb()`

(OK Callback)
Executed when the user clicks the "OK" button. This function internally calls `apply_cb()` , thereby performing the attribute assignment and then closing the dialog window.

`GetUnloadOption()`

Defines how the program is handled in NX memory after execution. Here, `Session.LibraryUnloadOption.Immediately` is set. This means that memory is released immediately after the script has been executed. This prevents memory leaks during an NX session.

The User Interface (`add_PartID.dlx`)

The included XML file (`.dlx`) defines the visual appearance of the NX dialog. It includes:

- The window title: **"Add Part ID"**
- A group (`group0`) for visual structuring
- The essential text input field (`string0`) labeled **"Enter Part ID"**.

Installation & Execution in NX

1. Download the files `add_PartID.cs` and `add_PartID.dlx`
2. Ensure that both files are located in the same directory or placed according to the NX environment variables (e.g., in the `application` folder under `UGII_USER_DIR`)
3. Open a part in Siemens NX.
4. Execute the C# file via (File -> Execute -> NX Open) or integrate it as a button into your NX user interface (Ribbon).

Troubleshooting

- **Error during startup (dialog cannot be found):** By default, the C# script searches for the file `C:\Users\XXXX\XXXXX\add_PartID.dlx` (as defined in the code). If the file has been moved, the path in the variable `theDlxFileName` in the C# code must be adjusted, or the `.dlx` file must be placed in a standardized NX folder ($UGII_USER_DIR/application/) so that NX can find it automatically.
- **Message "No active part available":** The script was executed without a part being open, or the displayed part is not the "Work Part". Set the part as the Work Part by right-clicking it in the Assembly Navigator and try again.

05 NX Sync Properties with AAS

Text

06 Attribute Export Skript

This Python script uses the NXOpen Python API to read specific attributes of an open NX part and export them into a CSV file.

The CSV file contains selected mass properties of a part, e.g.:

- Volume
- Mass
- Surface area

The file is automatically saved in the same directory as the NX part.

Basic Functionality

The script executes the following process:

1. Establish a connection to the current NX session
2. Access the active work part
3. Define the attributes to be read
4. Iterate over all attributes of the part
5. Extract the required values
6. Write the data to a CSV file

Code Structure and Functions

main()

The entry point of the script.

Functions:

- Access the current NX session
- Access the active work part
- Define the required attributes
- Read all part attributes
- Filter the relevant attributes
- Export the values to a CSV file

Definition of the required attributes

The following attributes are exported:

```
wanted_attributes = [
"MassPropVolume",
"MassPropMass",
"MassPropArea"
]
```

These attributes correspond to mass properties of an NX part.

Attribut	Description
MassPropVolume	Volume of the part
MassPropMass	Mass of the part
MassPropArea	Surface area of the part

Creation of the export file path

The file path of the CSV file is generated automatically:

Installation & Execution in NX

Ensure that Siemens NX with the NX Open Python API is installed. Save the script in an NX script directory or a project folder. Open a part in Siemens NX. Execute the script via the following menu:

- File → Execute → NX Open
- Select the Python script
- After execution, a CSV file is automatically generated

Troubleshooting (Fehlerbehebung)

No CSV file is generated
Possible causes:

- No active work part is available
- The part does not contain the required attributes
- Missing write permissions in the target directory (CSV File already open)

Attributes are not found
Possible causes:

- Attributes have different names
- Attributes are not stored as user attributes
- Alias and title differ

07 Recycling NX R-BOM to AAS

Vergleich von CAD-Systemen im Kontext der Verwaltungsschale (AAS)

Wenn sich im Rahmen der Entwicklung die Frage stellt, mit welchem CAD-System gearbeitet werden soll, kann auf praktische Erfahrungen aus unterschiedlichen Projekten zurückgegriffen werden. Wie in der nachfolgenden Tabelle dargestellt, unterscheiden sich die drei etablierten CAD-Systeme Siemens NX, PTC Creo und Autodesk Fusion 360 insbesondere in den Punkten Lizenzen, Funktionsumfang und Benutzerfreundlichkeit voneinander.

Kriterium	**Siemens NX**	**PTC Creo**	**Autodesk Fusion 360**
Lizenzen	0 Enterprise-Lizenz erforderlich zum Arbeiten mit Journals	[+/-/0] Begründung...	🚧 Work in Progress
Funktionsumfang	+ Sehr groß, alles was über Python abgedeckt werden kann	[+/-/0] Begründung...	🚧 Work in Progress
Benutzerfreundlichkeit	+ Durch Integration in die NX-Oberfläche	[+/-/0] Begründung...	🚧 Work in Progress

Tabelle: Vergleich ausgewählter CAD-Systeme für die AAS-Integration

Prerequisites

Für die erfolgreiche Einrichtung und Ausführung der Integration zwischen Siemens NX und der AAS müssen folgende Systemvoraussetzungen erfüllt sein:

Software & Lizenzen

- [ ] **Siemens NX:** Lokale Installation (z. B. Version 2021).

- [ ] **NX-Lizenz:** Enterprise- bzw. NX Open-Lizenz zur Ausführung und Kompilierung von Skripten und externen Programmen.
- [ ] **AASX Package Explorer:** Tool zur Betrachtung und Validierung der Verwaltungsschalen.

Entwicklungsumgebung & Sprachen

- [ ] **Python:** Eigenständige Installation (ab Version 3.11) für Adapter weil in der NX 2021 die Python Version nur die (Version 3.8 ist und in dieser die lxml-Bibliothek nicht enthalten ist).
- [ ] **IDE:** Visual Studio (für z.B. C#) oder Visual Studio Code (für Python) zum anpassen der Sktipte.

Systemrechte & Konfiguration

- [ ] **Administratorrechte:** Lokale Windows-Adminrechte (für Installationen via `pip` und das Setzen von Umgebungsvariablen).
- [ ] **NX-Rolle:** Eine fortgeschrittene Benutzerrolle (z. B. "Advanced"), um den Entwickler-Tab in der NX-Oberfläche freizuschalten.

Faziz and Ausblick

Hilfsmittel und Quellen

Eingesetzte Hilfsmittel

Folgende Tools haben bei der Umsetzung und Dokumentation des Projekts unterstützt:

- **DeepLcom:** Für Übersetzungen von Deutsch in Englisch.
- **Gemini:** Als KI-Assistent für die Markdown-Formatierung und Code-Snippets.
- **Drawio:** Zur Erstellung der Diagramme.
- **Visual Studio:** Als Entwicklungsumgebung für die Programmierung.
- **GitHub:** Für die Versionsverwaltung und Veröffentlichung des Codes.
- **Siemens NX:** Für die CAD Arbeiten.

Quellen

Die theoretische und technische Basis dieses Projekts stützt sich auf folgende Dokumente:

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[2] European Commission, "Commission Regulation (EU) 2019/2022 of 1 October 2019 laying down ecodesign requirements for household dishwashers," *Official Journal of the European Union*, L 315, pp. 267-284, Dec. 2019.

[3] European Parliament and Council of the European Union, "Regulation (EU) 2024/1781 of 13 June 2024 establishing a framework for the setting of ecodesign requirements for sustainable products," *Official Journal of the European Union*, L, Jun. 2024.

[4] European Parliament and Council of the European Union, "Regulation (EU) 2023/1542 of 12 July 2023 concerning batteries and waste batteries," *Official Journal of the European Union*, L 191, pp. 1-117, Jul. 2023.

[5] European Parliament and Council of the European Union, "Directive 2012/19/EU of 4 July 2012 on waste electrical and electronic equipment (WEEE)," *Official Journal of the European Union*, L 197, pp. 38-71, Jul. 2012.

[6] European Parliament and Council of the European Union, "Directive 2008/98/EC of 19 November 2008 on waste and repealing certain Directives," *Official Journal of the European Union*, L 312, pp. 3-30, Nov. 2008.

[7] European Parliament and Council of the European Union, "Regulation (EC) No 1907/2006 of 18 December 2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH)," *Official Journal of the European Union*, L 396, pp. 1-849, Dec. 2006.

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Siemens NX 2021 UI-Styler

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